National University of Modern Languages

Unity Task 2

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Course:	**Computer Graphics**
Class:	**BSCS VII-A**
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1. Objective

To enhance the previously created Basic3DScene by integrating additional 3D objects, particle effects, and other interactive or visual elements to better understand Unity's core features including materials, lighting, animation, and particle systems.

2. Tools and Environment

Software Used: Unity Editor version 6.1.0 (6000.1.5f1)

Target Platform: Windows, Mac, Linux

Development Mode: 3D Project Template

3. Implantation Overview

3.1 Project Setup

- The existing project Basic3DScene was loaded via Unity Hub.
- A new scene setup was continued within SampleScene.

3.2 Added 3D Objects

- Additional primitives such as:
    - **Capsule** (Capsule) – with a pink material
    - **Sphere** (Sphere) – with a blue material
    - Another **Cube** (named Cube (1)) – to diversify geometry
- All objects were positioned and rotated uniquely to create a visually balanced composition.

3.3 Materials and Textures

- New materials were created:
    - **Pink (Capsule)**
    - **Red (Cube)**
    - **Dark Gray (Ground)**
    - **Yellow (Lighting object / placeholder)**
    - **Blue (Sphere)**

- Materials were assigned via the Inspector, as seen in the Project > Assets > Material folder.

3.4 Particle System (Fire Effect)

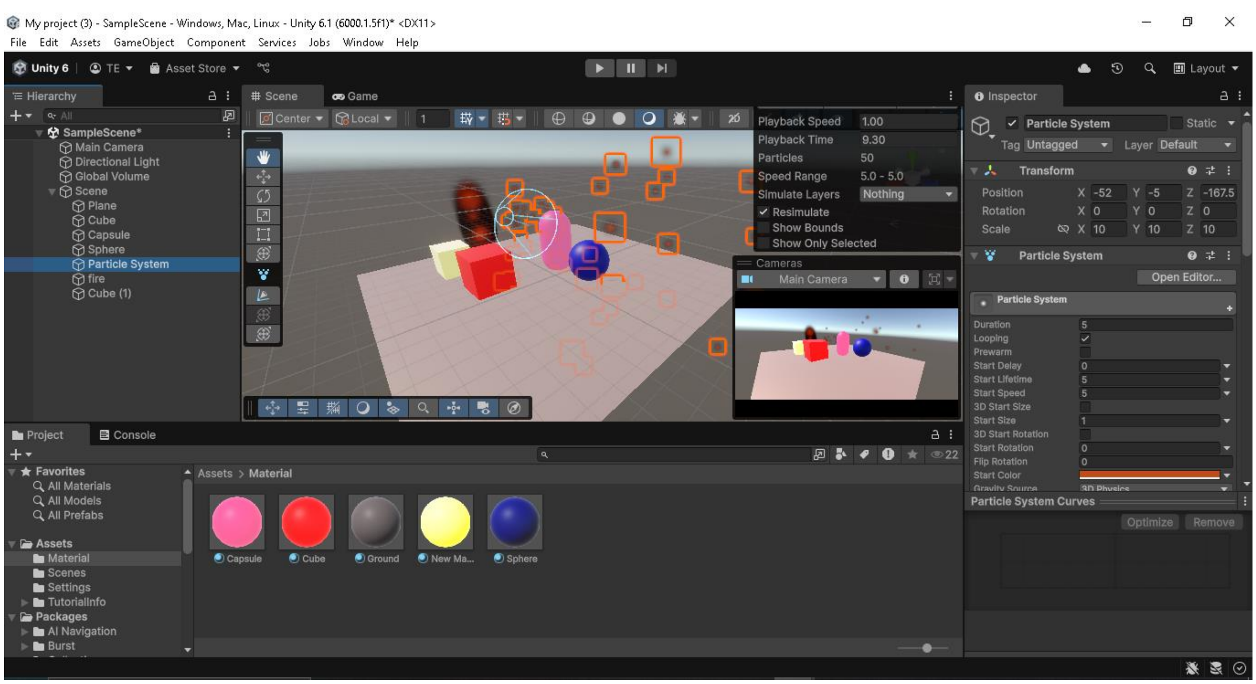
- A **Particle System** GameObject was added to simulate fire or spark effects.
- The system was:
  - Positioned near the red cube.
  - Customized with orange color, start lifetime of 5s, looping enabled, and size modifications.
  - Enhanced visual appeal and added dynamics to the scene.

3.5 Lighting and Camera

- A **Directional Light** was used to maintain global illumination.
- **Main Camera** repositioned and previewed in real-time to ensure good visibility of all objects and the fire effects.

4. Screenshot

Below is a screenshot showing the objects



5. Challenges and Learning Outcomes

- Particle system tweaking was a challenge, particularly adjusting parameters to make the fire effect realistic.
- Improved understanding of:
  - Scene composition and hierarchy organization
  - Custom materials and object aesthetics
  - Transformations and camera setup
- Learned to debug visual inconsistencies using the Scene view and Game view together.

6. Conclusion

This lab task successfully extended the Basic 3D Scene with multiple objects, unique materials, and a particle effect (fire sparks). It demonstrates progress in mastering Unity's editor tools and preparing for more advanced features like interactivity, animation, and physics in future tasks.